

7th Sem

-6 NOV 2024

Reg. No.: _____

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B.Tech Missed Mid Semester Examination – November 2024

Seventh Semester CSE

Computer Science and Engineering

19CSE353 Mining of Massive Datasets

Duration : Two hours

Maximum : 50 Marks

CO	Course Outcomes
CO1	Understand the importance of data to business decisions, strategy and behavior. Predictive analytics, data mining and machine learning as tools give new methods for analyzing massive data sets.
CO2	Expose the students to big data systems like Hadoop, Spark and Hive.
CO3	Explore means to deal with huge document databases and infinite streams of data to mining large social networks and web graphs. Also, learn Algorithms suitable for large scale mining.
CO4	Use of analytic tools, case-studies to provide firsthand insight into how big-data problems and their solutions are commercially derived.
CO5	Design Large Scale Machine Learning algorithms with Practical hands-on experience for analyzing very large amounts of data.

Answer all the questions

- Given two sets, $A=\{1, 2, 3, 4, 5\}$ and $B=\{3, 4, 5, 6, 7\}$, apply the MinHash technique using the following three hash functions: [CO1]
[BTL3]
 - $h1(x)=(2x+1) \bmod 7$
 - $h2(x)=(3x+2) \bmod 7$
 - $h3(x)=(5x+1) \bmod 7$
 - Compute the MinHash signatures for both sets using each hash function. [4]
 - Estimate Jaccard similarity of sets A and B using the MinHash signatures. [1]
 - Compare the results obtained from options (a) and (b). [1]
- Calculate the PageRank values for each page in Fig. 1 using two iterations of the PageRank algorithm, assuming a damping factor of 0.85. Explain how the damping factor affects the PageRank results. [5] [CO3]
[BTL3]

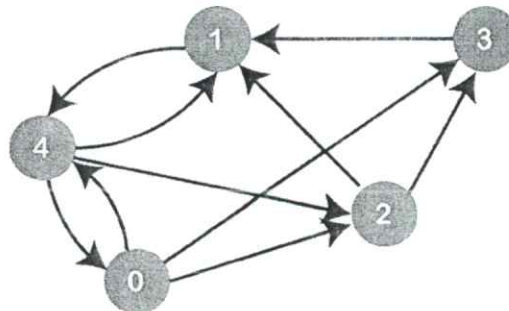


Fig. 1

3. How are failures of the Mapper, Reducer, and Master tasks handled during a MapReduce job? [3] [CO2]
[BTL1]
4. How does the stream data model facilitate real-time processing and analysis of continuously generated data, particularly in the context of social media, such as in real-time sentiment analysis during a product launch? Explain the same in detail. [4] [CO3]
[BTL2]
5. The database given below in Table 1 has five transactions from five customers. Using *Apriori* algorithm, find the association rules in this set, if the minimum support (i.e. coverage) is 60% and the minimum confidence (i.e. accuracy) is 80%. [Note: minimum support threshold is 60% of the total transactions] [6] [CO3]
[BTL3]

Table 1

Customer	Items
1	Orange Juice, Soda
2	Milk, Orange Juice, Window Cleaner
3	Orange Juice, Detergent
4	Orange Juice, Detergent, Soda
5	Window Cleaner, Soda

6. Suppose that the task is to cluster points (with (x, y) representing location) into two clusters, where the points are $A_1(2,10)$, $A_2(2,5)$, $A_3(8,4)$, $B_1(5,8)$, $B_2(7,5)$, $B_3(6,4)$. The distance function is Euclidean distance. Suppose initially A_1 , and B_1 are the center of each cluster, respectively. Use the *k-means* algorithm to show the final two clusters or two more iterations, whichever occurs first. [5] [CO3]
[BTL3]
7. Clustering Using REpresentatives (CURE) algorithm can find arbitrary shaped cluster while other Hierarchical approaches cannot handle such shapes. Justify this statement. [3] [CO3]
[BTL2]
8. Imagine you're tasked with analyzing customer purchasing behavior for an online retail store. The dataset contains attributes like purchase frequency, total spending, and time since the last purchase. Your goal is to segment customers into distinct groups for targeted marketing. Given the options of K-means, Hierarchical clustering, and DBSCAN, which algorithm would you choose, and why? Consider how each algorithm addresses challenges such as an unknown number of clusters, varying cluster densities, and the presence of outliers. [4] [CO3]
[BTL4]
9. Find the Singular Value decomposition of the matrix given below. Show the work. [6] [CO1]
[BTL3]

$$M = \begin{bmatrix} 1 & 1 \\ 2 & 2 \\ 2 & 2 \end{bmatrix}$$

10. Use *complete* link agglomerative clustering to group the data described by the distance matrix listed in Table 2. Show the dendrogram. [5] [CO3] [BTL3]

Table 2

	A	B	C	D
A	0	1	4	5
B		0	2	6
C			0	3
D				0

11. The measurements are made on seven individuals in a random sample from a population (Table 3). The standardized dataset is given in Table 4 and the generated covariance matrix is given in Table 5. Identify the principal components of the sample and hence derive the new data set. Reduce the dimension of the sample data set if the contribution of the least significant component is less than 15%. [3] [CO1] [BTL3]

Table 3

X	1	1.5	3	5	3.5	4.5	3.5
Y	1	2	4	7	5	5	4.5

Table 4

X	-1.46	-1.12	-0.10	1.27	0.24	0.93	0.24
Y	-1.53	-1.03	-0.04	1.46	0.46	0.46	0.21

Table 5

1.00	0.97
0.97	1.00

Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	15	BTL 1	3
CO2	3	BTL 2	7
CO3	32	BTL 3	36
CO4	0	BTL 4	4
CO5	0	BTL 5	0
		BTL 6	0